Test Intelligent Systems - DigiTenT/NU

Number of questions: 33
Session period: 22 november 2024 15:30 - 18:45
Duration: —

Instruction

Exam duration: 2 HOURS 15 MINUTES (135 minutes) unless you get extra time.

The following tools are permitted:
- Calculator
- Scrap paper (nothing written or printed on it)
- Check here for the formula sheet

This first partial exam consists of 33 questions of varying complexity. In total you can reach maximally: 120 Points. This means that you have roughly a minute per point.

There are 6 sections with material from the first 7 lectures (sets/graphs/tree, state-spaces, uninformed, informed and adversarial search, and logical agents and knowledge graphs). There are some simple reproduction questions (2-3 points), understanding questions (3 to 4 Points) and do-questions (4 to 8 Points).

Make sure you do not lose too much time on individual questions.

Multiple *Choice* questions have only **one** correct solution. In this case, the options are given by circles. **Multiple *Answers*** questions have one or more correct solution. In this case, the options are given by squares.

The score of Multiple Answer questions is calculated automatically by the system as follows: Score = proportion good * (1 - proportion wrong * (1/a + (((n-k)/n)*(1-1/a))), where n is the total number of alternatives, k the number of correct answers and a a weighting factor set to 2. The more correct answers you give the more points you get, but wrong answers also count negatively.

Set, Graphs and Trees

The following set of questions are about sets, graphs, and trees.

There are 3 questions for a total of 9 points.

Question 1 – Multiple choice – Question-ID: 456596 (2 points)

Given |U| = 30, |A| = 16, |B| = 13, and |A ∩ B| = 4, what is the cardinality of the complement of A union B, formally written as (A ∪ B)'?

A 0

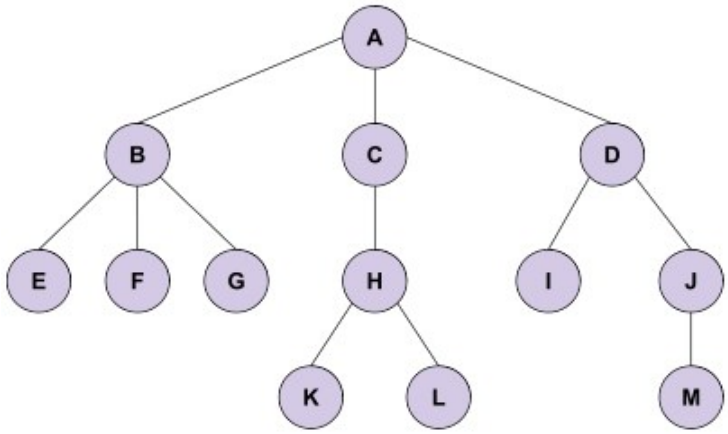
B 4

C 1

D 5

Question 2 – Multiple response – Question-ID: 456598 (3 points)

Based on the following tree, which statements are correct?



A The height of node H is 1.

B Node J is a leaf.

C Node F is a leaf.

D Node J can be identified uniquely as the child of their parent node.

E Node M can be identified uniquely as the child of their parent node.

F The height of node H is 2.

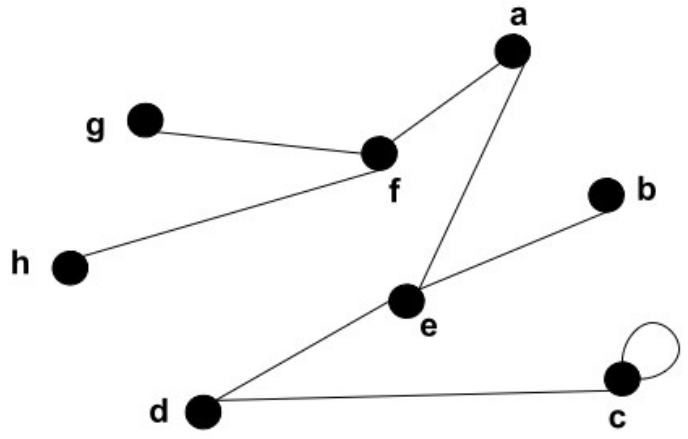
G The ancestors of node K are nodes A, C and H.

H The descendents of node K are nodes A, C and H.

I The tree has depth 4.

Question 3 – Multiple choice – Question-ID: 456601 (4 points)

Choose the adjacency matrix that correctly represents the following graph.


a
g
f
b
h
e
d
c

A

	a	b	c	d	e	f	g	h
a	0	0	0	0	1	1	0	0
b	0	0	0	0	1	0	0	0
c	0	0	0	1	0	0	0	0
d	0	0	1	0	1	0	0	0
e	1	1	0	1	0	0	0	0
f	1	0	0	0	0	0	1	1
g	0	0	0	0	0	1	0	0
h	0	0	0	0	0	1	0	0

B

	a	b	c	d	e	f	g	h
a	0	0	0	0	0	1	0	0
b	0	0	0	1	1	0	0	0
c	0	0	1	1	0	0	0	0
d	0	1	1	0	1	0	0	0
e	0	1	0	1	1	0	1	0
f	1	0	0	0	0	0	1	1
g	0	0	0	0	1	1	1	0
h	0	0	0	0	0	1	0	1

C

	a	b	c	d	e	f	g	h
a	0	0	0	0	1	1	0	0
b	0	0	0	0	1	0	0	0
c	0	0	1	1	0	0	0	0
d	0	0	1	0	1	0	0	0
e	1	1	0	1	0	0	0	0
f	1	0	0	0	0	0	1	1
g	0	0	0	0	0	1	0	0
h	0	0	0	0	0	1	0	0

D

	a	b	c	d	e	f	g	h
a	0	0	0	0	1	1	0	0
b	0	0	0	0	1	0	0	0
c	0	0	1	1	0	0	0	0
d	0	0	0	0	1	0	0	0
e	0	0	0	0	0	0	0	0
f	0	0	0	0	0	0	1	1
g	0	0	0	0	0	0	0	0
h	0	0	0	0	0	0	0	0

Questions on Uninformed Search

The following set of questions are about uninformed search.

There are 7 questions for a total of 22 points.

Question 4 – Multiple response – Question-ID: 284562 (3 points)

Which of the following statements about state-space search are correct? More than one answer may be correct.

- **A** The best sequence that leads to the goal is chosen from the possible sequence of transitions.
- **B** The search starts in the initial state.
- **C** The set of all possible states is referred to as a State Space.
- **D** The search ends whenever a leaf node is reached (so no successors)
- **E** The cost is the same for every action.

Question 5 – Multiple choice – Question-ID: 395691 (2 points)

When is a node considered visited while searching a tree?

- **A** When a node is removed from the fringe.
- **B** When a node is put into the fringe.
- **C** When a node is replaced by its children.
- **D** When its parent node is removed from the fringe.

Question 6 – Multiple response – Question-ID: 343919 (4 points)

Which option(s) is (are) true about uninformed search? There may be more than one correct answer.

- **A** As Depth-First Iterative Deepening is a variant of Depth-First Search, there are also cases in which it does not terminate.
- **B** If a solution exists, breadth-first search is guaranteed to find it unless the branching factor is infinite.
- **C** Depth-First Iterative Deepening has an exponentially higher space complexity than Breadth-First Search.
- **D** Depth-First Search always uses less memory than Breadth-First Search, whatever the search tree looks like.
- **E** Depth-First Iterative Deepening overcomes one of the most important limitations of Depth-First Search, as it can avoid getting stuck in loops.

Question 7 – Multiple choice – Question-ID: 284682 (2 points)

Not all algorithms are optimal in the sense of finding the best solution if it exists.
When is it considered **optimal** to choose Breadth-First Search?

- **A** The number of nodes is small
- **B** Costs are not equal for every step
- **C** Costs are equal for all steps
- **D** The branching factor is infinite

Question 8 – Multiple choice – Question-ID: 457603 (3 points)

Imagine having to search through the state-space for the following real world problems. Which statement about approaches to use in order to solve these real-world problems is true?

A Analysing MRI data to provide support to doctors performing a medical diagnosis should be done in a data driven way

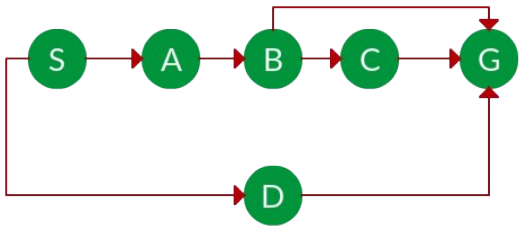
B SAT solving should be done in a goal driven way

C Providing bio-feedback (heart rate, oxygen level, etc.) for scuba divers should be done in a goal driven way.

Question 9 – Multiple choice – Question-ID: 457593 (4 points)

To reach goal state G from the initial state S, which order of node-exploration would follow for Breadth-first search (BFS) on the below State-Space?

Nodes are added to the fringe in alphabetical order (i.e. two children X and Y are added in the order X,Y to the fringe).



A S->A->D->B->G

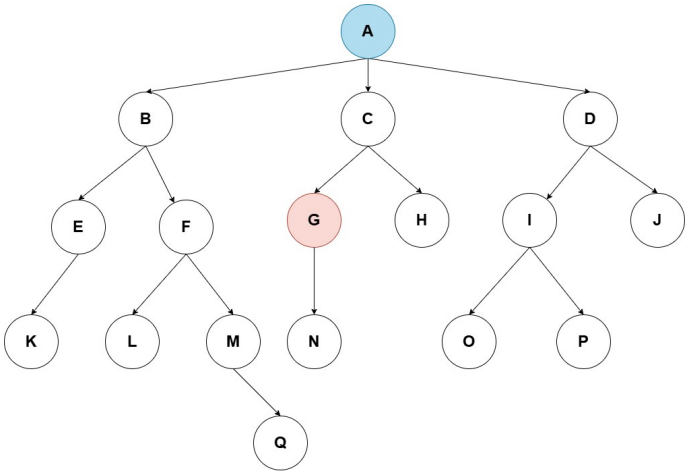
B S ->A-> D -> G

C S -> A -> B -> C -> G

D S -> D -> G

Question 10 – Fill in (multiple) – Question-ID: 457601 (4 points)

Use the following tree with initial state A and goal state G to answer the question about uninformed search.
Assume that the tree-search method you're applying adds nodes to the fringe in **alphabetical** order. In other words, the children B,C of A are added to the fringe as B,C.



When applying Depth-First Search to the tree, which node is removed from the fringe next after F? _____

When applying Depth-First Search, which node is evaluated next after L? _____

How many nodes are visited before you find the goal state **G** when applying Breadth-First Search? _____

Questions on Informed Search

The following set of questions are about informed search.

There are 7 questions for a total of 27 points.

Question 11 – Multiple choice – Question-ID: 457831 (2 points)

The term Euclidean distance refers to the straight-line distance between two points, as if you could move directly from one point to another in a straight path. Formally, it is the square root of the sum of the squared horizontal and vertical distances between points.

Why is the Euclidean distance an admissable heuristic?

A Because the Euclidean distance provides an exact measure of the path cost in all grid-based environments.

B Because the predicted distance from one point to another is an underestimate of the real path, ensuring that the heuristic is pessimistic.

C Because Euclidean distance is the shortest possible path between two points, making it a good lower bound for actual travel distance on a grid.

D Because the Euclidean distance is always an overestimate of the actual path cost between two points.

Question 12 – Multiple response – Question-ID: 457489 (3 points)

Which of the following search methods are incomplete, meaning they may fail to find a solution even if one exists?

- **A** A-search
- **B** Hill-climbing
- **C** Greedy best-first search
- **D** Depth-first iterative deepening
- **E** Breadth-first search
- **F** Depth-first search
- **G** Beam search
- **H** A*-search

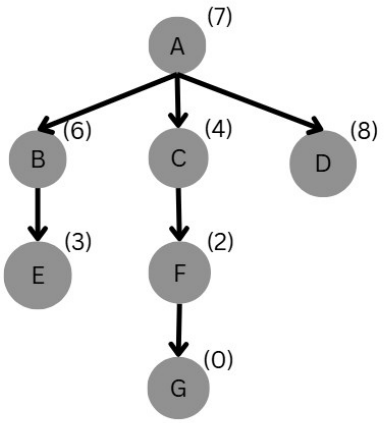
Question 13 – Multiple choice – Question-ID: 202980 (2 points)

Hill climbing always finds an optimal solution to a search problem, but it might take longer than eg. a more informed best-search strategy.

- **A** True
- **B** False

Question 14 – Fill in – Question-ID: 457642 (4 points)

Consider the following state-space, in which node G is the goal node. The lower the heuristic values, the better. The numbers next to the nodes represent the heuristic values.

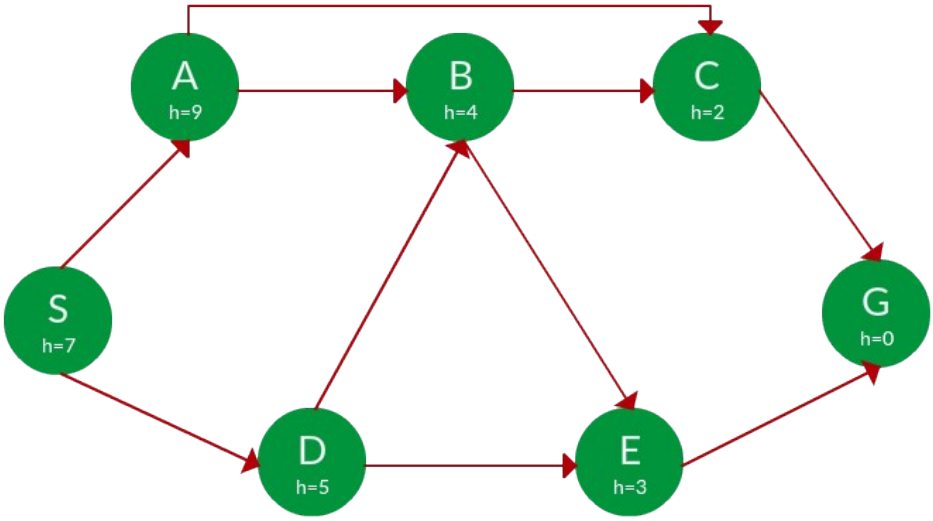


What is the order in which the nodes are searched by using Hill Climbing? Write down the letters of the nodes. Choose in alphabetical order when necessary.

Format your answer by separating nodes with comma's and NO spaces, i.e. X,Y,Z

Question 15 – Multiple choice – Question-ID: 284829 (4 points)

Find the path to reach goal state G starting from the start state S using greedy best-first search. (The heuristic values h of each node are mentioned on each of the nodes.)



A S -> D -> B -> E -> G

B S -> A -> C -> G

C S -> A -> B -> E -> G

D S -> D -> E -> G

E None of these paths.

F S -> A -> B -> C -> G

Question 16 – Multiple choice – Question-ID: 284825 (2 points)

The A* approach evaluation function *f(n)* consists of:

A Cost of the path from the start node to current node *g(n)* + heuristic function *(h(n)*

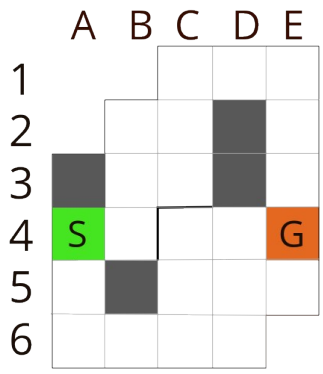
B The mean of the cost of the path from the start node to the current node *g(n)* + heuristic function *h(n)*

C The cost of the path from the start node to the current node *g(n)*

D The maximal value of the heuristic function at the current node *h(n)*

Question 17 – Multiple choice – Question-ID: 233370 (4 points)

Consider the following maze, in which A4 is the starting point and E4 the goal. The dark squares are not accessible to the algorithm, and the black lines are walls that cannot be crossed.
What is the correct order in which nodes are visited when applying A* with Manhattan distance as h(n)? This means that h(n) is an addition of the horizontal and vertical distances, without taking account the walls. Use alphabetical and numerical order when you encounter equal values.



A A4, B4, B3, C3, A5, A6, B6, C6, C5, D5, D4, E4

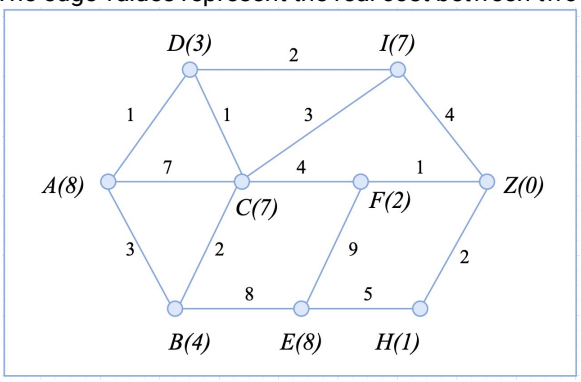
B A4, A5, A6, B6, C6, D5, E5, E4

C A4, B4, A5, B3, C3, A6, B2, B6, C2, C6, C5, C4, D4, E4

D B4, B3, B2, C2, C1, D1, E1, A5, A6, B6, C6, D6, D5, D4, E4

Question 18 – Fill in (multiple) – Question-ID: 343912 (6 points)

Consider the following graph. The value in parentheses represents a heuristic function that estimates the distance to the goal. The edge values represent the real cost between two nodes.



Describe the path which the A/A* search algorithm will take to reach node Z from node A.
Format your answer by separating nodes with comma's and NO spaces, i.e. X,Y,Z

Is the algorithm you applied in the previous step A search or A* search?
Your answer should look like: A OR A*

Questions on Adversarial Search

The following set of questions are about adversarial search.

There are 4 questions for a total of 20 points.

Question 19 – Multiple choice – Question-ID: 202985 (2 points)

Which of the following statements about the Minimax algorithm is true?

A In a Minimax algorithm MAX tries to maximize the sum of all the possible moves MIN could take.

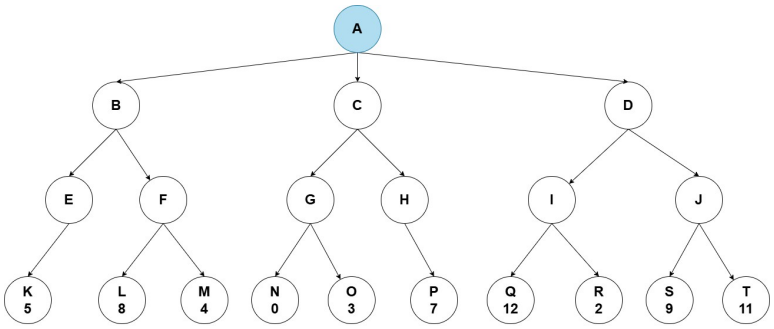
B In a Minimax algorithm MAX tries to minimize the worst-case outcome of all the possible moves MIN could take.

C In a Minimax algorithm MAX tries to maximize the worst-case outcome of all the possible moves MIN could take.

D In a Minimax algorithm MAX tries to minimize the sum of all the possible moves MIN could take.

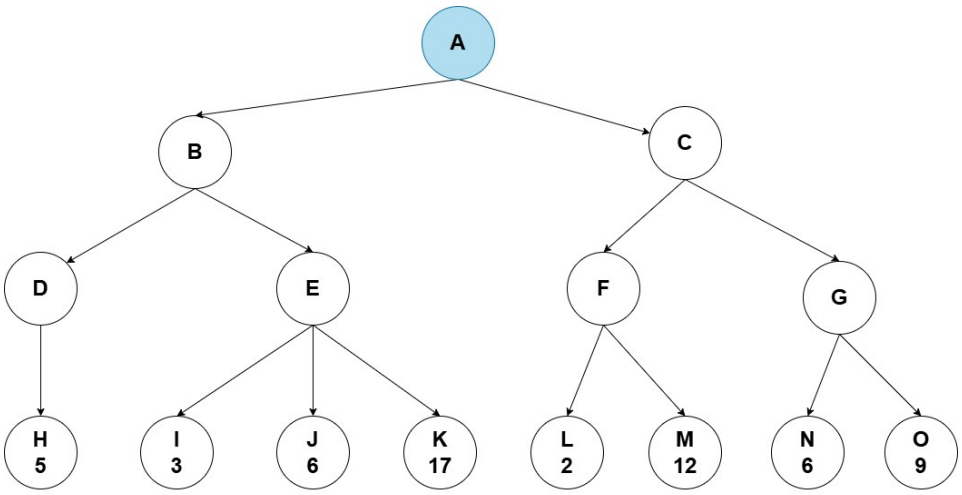
Question 20 – Fill in (numerical) – Question-ID: 457679 (4 points)

Assume that MAX makes the first move. Given the following tree, what is the value Minimax calculates for node A?



Question 21 – Multiple response – Question-ID: 457690 (6 points)

Consider the game tree given below when applying the Minimax algorithm.
Which of the following statements is/are true?



A When MAX starts, the value for A will be 17

B The value of node B is always 5, no matter if MIN or MAX starts

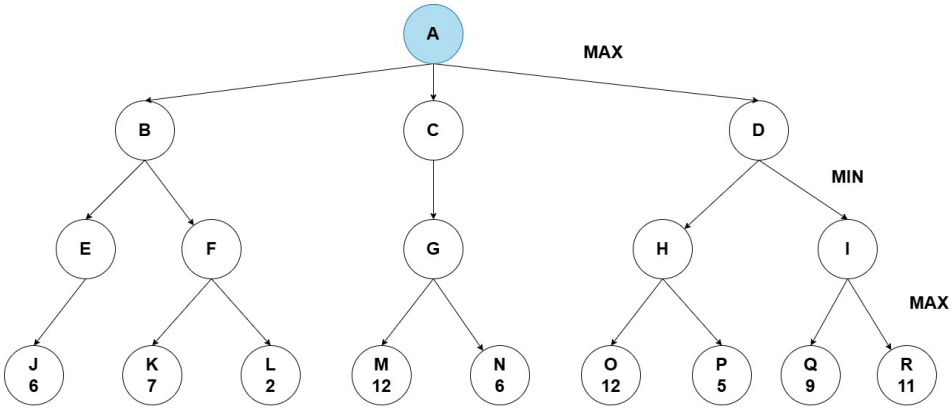
C The value of C is always 6, no matter if MIN or MAX starts

D When MIN starts, the value for A will be 5

E The value of node C is 2 when MIN starts

Question 22 – Multiple response – Question-ID: 457688 (8 points)

Consider the following Adversarial Search Space, where a Max player starts the game.



If you apply the Minimax algorithm with Alpha Beta Pruning, which nodes in the tree can you ignore (prune)? More than one answer can be correct.

A N

B I, Q, R

C L, N, P, R

D P

E C, G, M, N

F F, K, L

G L

Questions on Logical Agents

The following set of questions are about logical agents.

There are 9 questions for a total of 32 points.

Question 23 – Match (even) – Question-ID: 456604 (2 points)

Match the terms on the right with their correct definitions on the left by dragging each term to the corresponding definition.

A An interpretation function of the variables, which evaluates all sentences in the knowledge base to true.

B There exists an interpretation in which the sentence is true.

C A set of sentences that are assumed to be true, represented in a curly bracket notation.

D The sentence is a tautology.

1 **Satisfiability**

2 **Model**

3 **Knowledge base**

4 **Validity**

Question 24 – Fill in (multiple) – Question-ID: 456606 (3 points)

Complete the following truth table with the letter **T** for True and **F** for False. Make sure to use **capital letters!**

A	B	C	A ∨ C	¬B	¬B →(A ∨ C)
T	T	T	T	F	T
T	T	F		F	T
T	F	T	T		T
T	F	F	T	T	T
F	T	T	T	F	T
F	T	F		F	
F	F	T	T	T	T
F	F	F	F	T	

Question 25 – Multiple choice – Question-ID: 456609 (3 points)

Consider the following propositional formula: **(¬p ∧ q)** ↔ **r** .

Which of the following interpretations is a model?

A v(p)=False, v(q)=False, v(r)=False

B v(p)=False, v(q)=False, v(r)=True

C v(p)=True, v(q)=True, v(r)=True

D v(p)=True, v(q)=False, v(r)=True

Question 26 – Multiple response – Question-ID: 456610 (2 points)

Which of the following is/are the correct name(s) for the form where a formula is a conjunction of disjunctions of literals?

A Literal Normal Form

B Conjunctive Normal Form

C Disjunctive Normal Form

D Clause Normal Form

Question 27 – Multiple response – Question-ID: 456612 (4 points)

Consider the following formulas:

$\neg (p \wedge q) \rightarrow r$
$p \rightarrow (q \vee r)$
$\neg p \leftrightarrow q$

The CNF of these formulas includes some of the following clauses. Which of these clauses are part of the CNF? (Please select multiple answers.)

A $\neg p \vee q \vee \neg r$

B $q \vee \neg r$

C $p \vee r$

D $q \vee r$

E $p \vee q$

F $\neg q \vee \neg p$

G $\neg p \vee q \vee r$

H $\neg q \vee p$

Question 28 – Multiple choice – Question-ID: 456613 (2 points)

Given the following knowledge base in CNF:

$\{\neg A \vee B, C \vee B, A \vee \neg B \vee C, B \vee \neg C, \neg B \vee \neg A\}$

Which of the following variables would be assigned a value first using the DPLL algorithm with the MOMs heuristic?

A A

B B

C C

D None of the above

Question 29 – Multiple choice – Question-ID: 456615 (3 points)

Which of the following best describes how the DPLL algorithm can be used to prove entailment of a formula by a knowledge base using refutation?

A The CNF of the formula is added to the CNF of the knowledge base. Then the DPLL algorithm is applied. If a model is found, the formula is not entailed by the knowledge base.

B The CNF of the negation of the formula is added to the CNF of the knowledge base. Then the DPLL algorithm is applied. If a model is found, the formula is entailed by the knowledge base.

C The DPLL algorithm is applied to the CNF of the formula. If a model is found, the formula is not entailed by the knowledge base.

D The CNF of the negation of the formula is added to the CNF of the knowledge base. Then the DPLL algorithm is applied. If the search is exhausted without a model found, the formula is entailed by the knowledge base.

Question 30 – Fill in (multiple) – Question-ID: 456617 (5 points)

To prove the satisfiability of a formula F, we can use the DPLL algorithm. Complete the steps of the DPLL algorithm by placing the correct phrase in each empty space.

The list of phrases is given below, but bear in mind that there are **more phrases than empty spaces**, so only use the ones that best complete the steps of the algorithm:

- F
- literal
- any clause
- False
- dpll(F, ¬P)
- ¬literal
- ¬F
- empty clause
- dpll(¬F, P)
- True

Please copy-paste any symbols that may be difficult to type (e.g., ¬). Make sure to not put any unnecessary spaces in your answer.

```
dpll(F, literal){
    remove clauses containing [          ]
    shorten clauses containing [          ]

    if (F contains no clauses)
        return True
    if (F contains [          ] )
        return False
    if (F contains a unit or pure L)
        return dpll(F, L)

    choose P in F;
        if ([          ] )
            return True;
        else
            return dpll(F, P);  }
```

Question 31 – Fill in (multiple) – Question-ID: 456619 (8 points)

Given following knowledge base:

{A ∨ B,
¬A ∨ B ∨ ¬C,
A ∨ ¬B ∨ ¬C,
A ∨ ¬B ∨ C,
¬B ∨ C}

Fill in the gaps to prove satisfiability or unsatisfiability with DPLL. If you have a choice, apply an **alphabetical order** (starting with A), and **assign False before True** to the variables. **Do not use any heuristics**. Please keep the clauses in the order given above and do not leave spaces before, between or after propositions. To indicate negation, either copy ¬ (the negation symbol), or use - (the dash symbol).

Assigning the value False to variable A results in the clause set:

Given that _____ is a unit clause, assign the value _____ to variable _____, which results in the clause set: _____

This is a contradiction, so we need to backtrack on _____ and assign it the value _____, which results in the clause set:

According to the rule described above, assign the value _____ to variable _____, which results in the single clause:

This means that the knowledge base is _____ (satisfiable or unsatisfiable).

In case the knowledge base is satisfiable, provide a model in the following line. Otherwise, enter XXX (no spaces) in each of the fields.

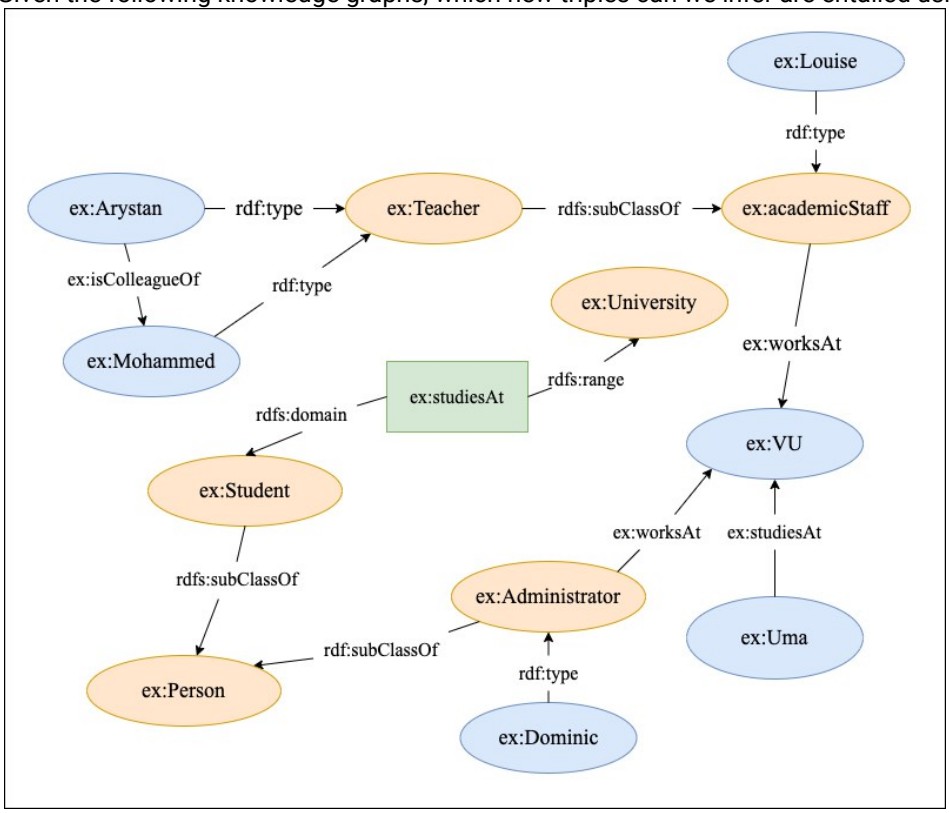
A= _____, B= _____, C= _____.

Knowledge Graph and Rulebased Reasoning

There are 2 questions with 10 points

Question 32 – Multiple response – Question-ID: 457639 (6 points)

Given the following knowledge graphs, which new triples can we infer are entailed using the RDFS inferencing rules?



A ex:Arystan ex:worksAt ex:VU

B ex:Mohammed rdf:subClassOf ex:academicStaff .

C ex:VU rdf:type ex:University

D ex:Arystan rdf:type ex:academicStaff .

E ex:Dominic rdf:type ex:Person

F ex:Mohammed rdf:type ex:Person

G ex:Person rdf:type ex:Teacher

H ex:Administrator ex:isColleagueOf ex:Teacher

Question 33 – Match (even) – Question-ID: 457640 (4 points)

Using Forward Checking rule-based reasoning, the following rewriting steps are taken to prove whether KB = {B → ¬(A ∨ ¬C), B} entails C.
Match each step on the left to the rule on the right that was applied in order to get to that step.

For the conclusion statement, match the correct entailment statement to the "Correct conclusion?" box and the incorrect entailment statement to "Incorrect conclusion?

A	¬(A ∨ ¬C)	**1**	C is **not entailed** by the knowledge base.
B	¬A ∧ C	**2**	C is **entailed** by the knowledge base.
C	¬A, C	**3**	And-elimination
D	Correct conclusion?	**4**	Modus Ponens
E	Incorrect conclusion?	**5**	De Morgan

End of test Intelligent Systems - DigiTenT/NU